The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Displacement-Body Quadratic Forcing Law
from Body-Lift Response Selection

Abstract

The current active-primary theorem already proves that the canonical minimal visible-response representative $\mathfrak{S}^{\mathrm{disp,resp,red,min}}$ and its visible-response operator family are canonical consequences of a smaller section-centered displacement-body quadratic forcing law on the fixed displacement body. The next honest benchmark-facing move is therefore no longer another same-body-law restatement, another same operator-extension relay, or another restored larger same-output package. It is to derive that current body-level forcing law itself from still more primitive explicit substrate structure.

The present note makes that move in the smallest explicit package presently available below the forcing-law frontier. For each sector it introduces a benchmark-faithful body-lift response selection: the fixed proto-section benchmark base and fixed section-centered displacement body together with a finite-dimensional selected substrate carrier, a smooth linear selected response lift over the displacement fibers, and a smooth positive self-adjoint selected response operator. The current displacement-body quadratic forcing law is then the selected substrate-response energy of the chosen response-carrying lift on the actual displacement body. Hence the current body-level forcing law is no longer primitive once this smaller explicit body-lift response selection is fixed.

The benchmark boundary remains unchanged throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. By the previous forcing-law theorem, the same canonical minimal visible-response representative, the same current body-response core, and every currently recorded downstream benchmark-facing consequence then follow unchanged. The remaining honest burden therefore moves below the current body-level forcing law to a deeper derivation or justification of the selected body-lift response package itself rather than another same-output replay.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has now spent the presentational freedom inside the current visible-response datum and, after that, the presentational freedom inside the current body-level forcing law itself. The visible-response operator family and canonical minimal visible-response representative are no longer primitive once the smaller section-centered displacement-body quadratic forcing law is fixed. The next honest question is therefore narrower again: what smaller explicit substrate structure forces that current body-level law?

The present note takes the smallest clean route currently available below the forcing-law frontier. It does not restore the full displacement-response substrate law, the larger projection/body presentation, the full canonical current-body image core, or any other larger same-output package as if those were again the live benchmark-facing datum. Instead it retains only a selected response-carrying substrate representative of each actual recorded displacement together with a positive substrate-response operator acting on that selected carrier. The current body-level forcing law is then read off as the selected substrate-response energy on the fixed displacement body.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim a completed first-principles derivation of GR from final substrate microphysics. Its narrower positive role is to show that the current section-centered displacement-body quadratic forcing law is already forced once the smaller selected body-lift response package is fixed.

2. Fixed Inputs from the Current Forcing-Law Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active route already fixes the following data and consequences.*

1. *the current proto-section benchmark base, in particular the recorded proto-section body* $\mathcal{B}_\sigma^{\mathrm{psec}}$ *and shell* $\mathcal{Z}_\sigma^{\mathrm{psec}}$*;*

2. *the fixed section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{ppprese c}}$$

whose fiber

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}}(y) \equiv \{v \in V_\sigma^{\mathrm{ppprese c}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}\}$$

is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*;*

3. *the recorded downstream continuation, namely*

$$\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,bodyresp}}, \quad \mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,bodyop}}, \quad \mathfrak{R}^{\mathrm{sub,proto,prese c,res}}, \quad \mathfrak{R}^{\mathrm{sub,proto,prese c}},$$
$$\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,op}}, \quad \mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb}}, \quad \mathfrak{R}^{\mathrm{sub,proto,prese c,sec}},$$

together with the current continuity-Hessian compressed core and benchmark one-metric observer package;

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Proposition 1 (What the current forcing-law frontier already proves)**.** *At current scope, the already-recorded forcing-law frontier theorems prove all of the following.*

1. *Any benchmark-faithful section-centered displacement-body quadratic forcing law canonically determines the visible-response operator family* $S_\sigma^{\mathrm{disp,resp,vis}}$ *and the canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *on the fixed displacement body.*

2. *Once that law is available, the current body-response core and all currently recorded downstream benchmark-facing consequences already follow unchanged from the same derived representative.*

Proof. Item (1) is exactly the content of the current theorem deriving the canonical minimal visible-response representative from the section-centered displacement-body quadratic forcing law. Item (2) is exactly the downstream factorization statement recorded there. □

3. Body-Lift Response Selection

Definition 3 (Benchmark-faithful body-lift response selection)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful body-lift response selection *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*

2. *A finite-dimensional Euclidean selected-lift carrier*

$$\mathcal{U}_\sigma^{\mathrm{disp,body,sel}}.$$

3. *A smooth family of linear selected response lifts*

$$\Lambda_\sigma^{\mathrm{disp,body,sel}}(y) : V_\sigma^{\mathrm{pppresec}} \to \mathcal{U}_\sigma^{\mathrm{disp,body,sel}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

used only on the recorded displacement-body fiber

$$\mathcal{D}_\sigma^{\mathrm{pppresec}}(y).$$

4. *A smooth family of positive self-adjoint selected response operators*

$$K_\sigma^{\mathrm{disp,body,sel}}(y) : \mathcal{U}_\sigma^{\mathrm{disp,body,sel}} \to \mathcal{U}_\sigma^{\mathrm{disp,body,sel}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

5. *Positivity on the actual displacement body:*

$$\left\langle \Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v, K_\sigma^{\mathrm{disp,body,sel}}(y)\Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v \right\rangle = 0 \qquad \Longleftrightarrow \qquad v = 0$$

for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ and every $v \in \mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$.

6. *Benchmark faithfulness in the sense of Definition 2(4).*

Remark 1. *Definition 3 is intentionally smaller than the older section-centered displacement-response substrate law and smaller than the later projection/body or current-image-core presentations. It retains only the explicit response-carrying selected substrate representative of each actual recorded displacement together with the positive substrate-response operator acting on that selected carrier. Ambient substrate-body points, projection/kernel decompositions, and other larger same-output presentations are not kept as separately primitive data here.*

4. Forced Section-Centered Displacement-Body Quadratic Forcing Law

Definition 4 (Selected-lift forcing energy)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ and a benchmark-faithful body-lift response selection. Define the selected-lift forcing energy on the recorded displacement body by*

$$\mathscr{F}_\sigma^{\mathrm{disp,resp,vis,body}}(y, v) \equiv \frac{1}{2} \left\langle \Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v, K_\sigma^{\mathrm{disp,body,sel}}(y)\Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v \right\rangle \tag{2}$$

for every $(y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}$.

Proposition 2 (The selected-lift forcing energy is a benchmark-faithful body-level quadratic forcing law)**.** *Fix a benchmark-faithful body-lift response selection for sector σ. Then the pair*

$$\left(\mathcal{D}_\sigma^{\mathrm{pppresec}}, \mathscr{F}_\sigma^{\mathrm{disp,resp,vis,body}}\right)$$

is a benchmark-faithful section-centered displacement-body quadratic forcing law on the fixed displacement body.

Proof. Because $y \mapsto \Lambda_\sigma^{\mathrm{disp,body,sel}}(y)$ and $y \mapsto K_\sigma^{\mathrm{disp,body,sel}}(y)$ are smooth, the forcing energy $\mathscr{F}_\sigma^{\mathrm{disp,resp,vis,body}}$ is smooth in y. Because $v \mapsto \Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v$ is linear on $V_\sigma^{\mathrm{pppresec}}$, the restriction of $\mathscr{F}_\sigma^{\mathrm{disp,resp,vis,body}}(y, \cdot)$ to the recorded displacement-body fiber $\mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$ is quadratic in v. Positivity and vanishing only at $v = 0$ on that fiber are exactly Definition 3(5). Benchmark faithfulness is

part of Definition 3(6). Therefore the induced pair satisfies the requirements of a benchmark-faithful section-centered displacement-body quadratic forcing law on the fixed displacement body. □

5. Forcing the Current Body-Law Frontier

Theorem 1 (Current displacement-body quadratic forcing law from body-lift response selection)**.** *Fix the primitive continuation data of Definition 1, the recorded benchmark base and downstream chain of Definition 2, and for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *a benchmark-faithful body-lift response selection in the sense of Definition 3. Then the current section-centered displacement-body quadratic forcing law is canonically forced by that smaller explicit package. More precisely:*

1. *the selected-lift forcing energy* $\mathscr{F}_\sigma^{\mathrm{disp,resp,vis,body}}$ *of Definition 4 is a benchmark-faithful section-centered displacement-body quadratic forcing law on the fixed displacement body by Proposition 2;*
2. *by Proposition 1(1), that law canonically determines the visible-response operator family* $S_\sigma^{\mathrm{disp,resp,vis}}$ *and the canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$*;*
3. *by Proposition 1(2), the current body-response core and every currently recorded downstream benchmark-facing consequence in Definition 2(3) then follow unchanged from the same derived representative;*
4. *the benchmark one-metric package* (1) *is preserved, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged;*
5. *consequently the remaining honest burden is no longer to restate the current section-centered displacement-body quadratic forcing law itself, but to derive or justify the smaller explicit body-lift response selection from still more primitive substrate structure.*

Proof. Item (1) is exactly Proposition 2. Item (2) is Proposition 1(1) applied to the law defined in (2). Item (3) is Proposition 1(2). Item (4) is the benchmark-faithfulness condition built into Definitions 2 and 3. Item (5) is the routing consequence of Items (1)–(4): once the current body-level forcing law is itself a canonical consequence of the smaller selected body-lift response package, another manuscript that merely restates that same law or merely replays its downstream consequences does not remove any remaining benchmark-facing freedom. □

Corollary 1 (The current body-level forcing law is no longer primitive)**.** *Under the hypotheses of Theorem 1, the current section-centered displacement-body quadratic forcing law is no longer a primitive stopping point on the recorded section-centered hidden-response route. It is a forced output of the benchmark-faithful body-lift response selection.*

Proof. Theorem 1(1) identifies the current body-level forcing law as the selected substrate-response energy of the smaller explicit package, while Theorem 1(2)–(3) identifies the same downstream benchmark-facing consequences. The law is therefore induced rather than postulated independently. □

6. Conclusion

The positive result of this note is a genuine derivational gain below the current displacement-body forcing-law frontier. The current section-centered displacement-body quadratic forcing law is no

longer primitive once the smaller body-lift response selection is fixed. The active line therefore no longer stops honestly at the body-law presentation itself.

The scope remains narrow and honest. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and nothing here upgrades adoption to a completed first-principles derivation of GR from final substrate microphysics.

The next open burden is therefore sharper again. The next benchmark-facing manuscript should derive or justify the current body-lift response selection itself from still more primitive explicit substrate structure. Another manuscript that merely restates the same selected-lift package, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, or another larger same-output current-image-core or reduced-carrier or projection/body or full substrate-law presentation is no longer the clean next target at current scope.